ML project Readme file

Description of Files in the ZIP file

The project ZIP file contains the following important files and documents:

1. **ML_PROJECT_DONE.ipynb** : original code of the project

2. **README.pdf** : to understand procedure

3. **Report File (PDF)**

4. **2 - Dataset named as `sp500_2011_2025.xlsx` and `new_df.xlsx`**

Project dependences table

Table 1: Project Dependencies and Purpose

Library/Module	Purpose
`numpy`	Numerical operations and array manipulation.
`pandas`	Data manipulation and analysis.
`matplotlib.pyplot, seaborn`	Data visualization (plotting).
`ta`	Calculation of technical analysis indicators (RSI, MACD, etc.).
`scikit-learn (sklearn)`	Machine learning framework for modeling and evaluation.
`sklearn.linear_model`	Linear models like Ridge, Lasso, and their cross-validated counterparts.
`sklearn.ensemble`	Ensemble models like RandomForestRegressor, GradientBoostingRegressor, and StackingRegressor.
`sklearn.model_selection`	Cross-validation (TimeSeriesSplit) and hyperparameter tuning (GridSearchCV, RandomizedSearchCV).
`sklearn.preprocessing`	Data scaling (StandardScaler).
`sklearn.metrics`	Model evaluation metrics (mean_squared_error, r2_score).
`joblib`	Saving and loading trained machine learning models.

Data Description

We downloaded data from Yahoo Finance from the date **03/01/2011 to 25/10/2025** as `SP500_2011_2025`.

For next prediction, data from **20/10/2025 to 31/10/2025** was downloaded as `new_df`.

After downloading the data, I made some changes in header part (names of columns) in that data from the Excel file that are:

from this

```
Price             Close          High           Low          Open      Volume
Date
2011-01-03  1271.869995   1276.170044   1257.619995   1257.619995  4286670000
2011-01-04  1270.199951   1274.119995   1262.660034   1272.949951  4796420000
```

to this

Date	Close	High	Low	Open	Volume
03-01-2011	1271.87	1276.17	1257.62	1257.62	4286670000
04-01-2011	1270.2	1274.12	1262.66	1272.95	4796420000

Instructions for Training and Inference

Firstly, we have taken the data, then cleaned the data and extracted features from it. After extracting features, we split the data into two parts: **Train Data** and **Test Data** based on date. The data till **30/04/2024** was used as the training data, and the data after that was used as the testing data.

Then, we trained our machine learning models such as `RandomForestRegressor`, `GradientBoostingRegressor`, `RidgeCV`, and `LassoCV` using the training data (including hyperparameter tuning for the Random Forest model).

Finally, we evaluated the fitted models on the test data to check their performance and accuracy.

After that, we created a model with the help of all four models, known as the **Stacking Ensemble**. This model combined the predictions of `RandomForestRegressor`, `GradientBoostingRegressor`, `RidgeCV`, and `LassoCV`, and it performed the best among all individual models.

Then, we used this Stacking Ensemble model to make predictions for a week volatility. After loading the new dataset, we compared the **predicted volatility** with the **actual volatility** to evaluate the model's performance.